

Component list for first iteration

Component	Description	Quantity	Electrical Specification	Size [SMD]	LCSC part#	Name	Component Link	Datasheet Link	Price x Qty
Capacitor	CVM1/MCU_V	x10	100nF 50V X7R ±10%	0603	C14663	CC0603KRX7R9BB104	C14663	C14663	\$0.0026 x 1
Capacitor	CVM2/VIN	x2	10uF 50V X7R ±10	1206	C89632	CL31B106KBHNNNE	C89632	C89632	\$0.0377 x 1
Capacitor	CCP/CAVVD/MCU_EN/2.7/1.2/VDD	x6	1uF 50V X7R ±10%	0805	C28323	CL21B105KBFNNNE	C28323	C28323	\$0.0099 x 1
Capacitor	MCU_V/SW_BK/3V3/3V3	x4	22uF 25V X5R ±20%	0805	C45783	CL21A226MAQNNNE	C45783	C45783	\$0.0308 x 1
Capacitor	CSOX	x3	22pF 50V C0G ±5%	0402	C1555	CL21A226MAQNNNE	C1555	C1555	\$0.0012 x 1
Capacitor	CFLY	x1	47nF 50V X7R ±10%	0603	C1622	CL10B473KB8NNNC	C1622	C1622	\$0.0052 x 1
Capacitor	SINP/SINNC/COSP/COSN	x4	68nF 50V C0G ±5%	1206	C77084	GRM31C5C1H683JA01L	C77084	C77084	\$0.1122 x 1
Capacitor	GMR/VDD/USBC/USBC_PD/DC	x5	4.7nF 50V X7R ±10%	0603	C1621	CL10B472KB8NNNC	C1621	C1621	\$0.0031 x 1
Resistor	RnFault	x1	5.1kΩ 50V 62.5mW Thick Film Resistor ±1%	0402	C105872	RC0402FR-075K1L	C105872	C105872	\$0.0006 x 1
Resistor	RBK/D+/D-/Diode blå-grønn	x3	22Ω 150V 125mW Thick Film Resistor ±1%	0805	C17561	0805W8F220JT5E	C17561	C17561	\$0.0024 x 1
Resistor	RSOX/IO0	x4	330Ω 150V 125mW Thick Film Resistor ±1%	0805	C17630	0805W8F3300T5E	C17630	C17630	\$0.0022 x 1
Resistor	G/FB/RESET	x3	100kΩ 150V 125mW thick Film Resistor ±1%	0805	C149504	0805W8F1003T5E	C149504	C149504	\$0.0023 x 1
Resistor	Enable, SINP/N, Cosp/N	x5	10kΩ 75V 100mW Thick Film Resistor ±1%	0603	C2906982	FRC0603F1002TS	C2906982	C2906982	\$0.0007 x 1
Resistor	FB/G	x2	22kΩ 150V 125mW Thick Film Resistor ±1%	0805	C17560	0805W8F2202T5E	C17560	C17560	\$0.0022 x 1
Resistor	VBUS_DISCH/DISCH	x2	1kΩ 150V 125mW Thick Film Resistor ±1%	0805	C95781	RC0805FR-071KL	C95781	C95781	\$0.0015 x 1
Resistor	EN	x1	220kΩ 150V 125mW Thick Film Resistor ±1%	0805	C17556	0805W8F2203T5E	C17556	C17556	\$0.0022 x 1
Resistor	Red LED	x1	75Ω 100mW 75V Thick Film Resistor ±1%	0603	C4275	0603WAF750JT5E	C4275	C4275	\$0.0015 x 1
Resistor	EN	x1	560kΩ 125mW 150V Thick Film Resistor ±1%	0805	C15785	0805W8F5603T5E	C15785	C15785	\$0.0021 x 1
Inductor	BOOT	x1	10uH 2A	1210	C39846661	XRTC322512S100MBCA	C39846661	C39846661	\$0.1224 x 1
LED	Red	x1	2.4V 20mA 40mW 615nm~630nm	0603	C2286	KT-0603R	C2286	C2286	\$0.0064 x 1
LED	Green	x1	2.6V~3.1V 5mA 100mW 513nm~528nm	0805	C2297	KT-0805G	C2297	C2297	\$0.0157 x 1
Connector	DC power	x1	2 Position Wire to Board Terminal Block	Through Hole 5.08mm	C8465	KANGNEX WJ500V-5.08-2P	C8465	C8465	\$0.1281 x 1
Connector	OUTPUT	x1	3 Position Wire to Board Terminal Block	Through Hole 5.08mm	C430602	DB128L-5.08-3P-BK-S	C430602	C430602	\$0.2321 x 1
Connector	USB-C Connector	x2	20V 5A -30°C~+80°C	8.94mmx7.35mmx3.26mm	C165948	TYPE-C-31-M-12	C165948	C165948	\$0.1810 x 1
Switch	EN, BOOT	x2	Black Round Button SMD	4mmx3mmx2mm	C720477	TS-1088-AR02016	C720477	C720477	\$0.0555 x 1
IC	USB-C PD	x1	3V/5.5V 160uA -40°C~+105°C	2.6mmx2.6mmx0.5mm	C2678061	STUSB4500QTR	C2678061	C2678061	\$1.3770 x 1
IC	Encoder	x1	3V~3.6V 7mA -40°C~+150°C	5mmx6mmx1.75mm	C2655191	TLE5009E1000FUMA1	C2655191	C2655191	\$3.7725 x 1
IC	Microcontroller	x1	2.4GHz 3V~3.6V 97mA -40°C~+85°C	20.5mmx15.4mmx2.4mm	C2913206	ESP32-S3-MINI-1-N8	C2913206	C2913206	\$4.5968 x 1
IC	Motor driver	x1	4.5-35V 8A peak 200-kHz PWM -40°C~+125°C	5mmx7mmx1mm	C20345369	DRV8316CQRGRGFRQ1	C20345369	C20345369	\$6.6228 x 1
IC	Voltage regulator	x1	500kHz Buck Adjustable 2A 4.5V~28V	1.6mmx2.9mmx1mm	C191884	TPS54202DDCR	C191884	C191884	\$0.2030 x 1
Mosfet	Power mosfet for power switching	x1	30V 40A	3.15mmx3.3mmx0.75mm	C49452333	STL9P3LLH6-TP	C49452333	C49452333	\$0.1990 x 1
Diode	VCC buss diode	x3	30V 5A	5mmx2.7mmx1mm	C50603	PMEG3050EP.115	C50603	C50603	\$0.4075 x 1
Diode	ESD Diode	x1	25VC Clamp ESD DIODE	1.6mmx3mmx1.4mm	C95343	ESDA25L	C95343	C95343	\$0.0826 x 1
Diode	ESD Diode	x3	38VC Clamp ESD DIODE	1.6mmx1mmx0.5mm	C19829649	ESDA25P35-1U1M	C19829649	C19829649	\$0.1098 x 1
Pin	Header pin	x12	3A 250V 2xpin header	3mm 2.54mm	C492401	PZ254V-11-02P	C492401	C492401	\$0.0084

Component list for second iteration

Component	Description	Quantity	Electrical Specification	Size [SMD]	LCSC part#	Name	Component Link	Datasheet Link	Price x Qty
Capacitor	CSOX	x3	22pF 50V COG ±5%	0402	C1555	0402CG220J500NT	C1555	C1555	\$0.0014 x 1
Capacitor	GMR/USBC/USBC_PD/DC	x4	4.7nF 50V X7R ±10%	0402	C1538	0402B472K500NT	C1538	C1538	\$ 0.0013 x 1
Capacitor	SINP/SINN/COSP/COSN/CLFY	x5	4.7nF 50V X7R ±10%	0603	C1622	CL10B473KB8NNNC	C1622	C1622	\$0.0057 x 1
Capacitor	BOOT/MCU/VCC/3V3/nFAULT/VDD	x9	100nF 50V X7R ±10%	0402	C307331	CL05B104KB54PNC	C307331	C307331	\$ 0.0047 x 1
Capacitor	CCP/CAVVD/MCU_EN/2.7/1.2/VDD	x6	1uF 50V X5R ±10%	0603	C15849	CL10A105KB8NNNC	C15849	C15849	\$ 0.0077 x 1
Capacitor	CVM2/VIN	x2	100uF 50V X5R ±10%	0805	C440198	GRM21BR61H106KE43L	C440198	C440198	\$ 0.0687 x 1
Capacitor	MCU_V/SW_BK/3V3/3V3	x4	22uF 6.3V X5R ±20%	0603	C59461	CL10A226M08NRNC	C59461	C59461	\$ 0.0101 x 1
Resistor	SW_BK	x1	22Ω 62.5mW Thick Film Resistor ±1%	0402	C25092	0402WGF220JTCE	C25092	C25092	\$ 0.0009 x 1
Resistor	MFET_FB/RED_DIODE	x2	100Ω 62.5mW Thick Film Resistor ±1%	0402	C25076	0402WGF1000TCE	C25076	C25076	\$ 0.0010 x 1
Resistor	RSOX/IO0	x4	330Ω 62.5mW Thick Film Resistor ±1%	0402	C25104	0402WGF3300TCE	C25104	C25104	\$ 0.0011 x 1
Resistor	VBUS_DISCH/DISCH	x2	1kΩ 100mW Thick Film Resistor ±1%	0603	C21190	0603WAF1001T5E	C21190	C21190	\$ 0.0016 x 1
Resistor	C+/C-/RnFAULT	x3	5.1kΩ 62.5mW Thick Film Resistor ±1%	0402	C25905	0402WGF5101TCE	C25905	C25905	\$ 0.0009 x 1
Resistor	MCU_Reset / SINP/N / Cosp/N	x5	10kΩ 62.5mW Thick Film Resistor ±1%	0402	C25744	0402WGF1002TCE	C25744	C25744	\$ 0.0013 x 1
Resistor	MFET_GATE/3V3_FB	x2	22kΩ 62.5mW Thick Film Resistor ±1%	0402	C25768	0402WGF2202TCE	C25768	C25768	\$ 0.0010 x 1
Resistor	MFET_GATE/3V3_FB/STUSB_Reset	x3	100kΩ 62.5mW Thick Film Resistor ±1%	0402	C25741	0402WGF1003TCE	C25741	C25741	\$ 0.0009 x 1
Resistor	3V3_Enable	x1	220kΩ 100mW Thick Film Resistor ±1%	0603	C22961	0603WAF2203T5E	C22961	C22961	\$ 0.0014 x 1
Resistor	3V3_Enable	x1	510kΩ 100mW Thick Film Resistor ±1%	0603	C23192	0603WAF5103T5E	C23192	C23192	\$ 0.0019 x 1
Resistor	3V3_Enable	x1	56kΩ 100mW Thick Film Resistor ±1%	0603	C23206	0603WAF5602T5E	C23206	C23206	\$ 0.0016 x 1
Inductor	BOOT	x1	10uH 2A	1210	C39846661	XRTC322512S100MBCA	C39846661	C39846661	\$0.1224 x 1
LED	Red	x1	2.4V 20mA 40mW 615nm~630nm	0603	C2286	KT-0603R	C2286	C2286	\$0.0064 x 1
LED	Grønn	x1	2.6V~3.1V 5mA 100mW 513nm~528nm	0805	C2297	KT-0805G	C2297	C2297	\$0.0157 x 1
Connector	USB-C Connector	x2	20V 5A -30°C~+80°C	8.94mmx7.35mmx3.26mm	C165948	TYPE-C-31-M-12	C165948	C165948	\$0.1810 x 1
Connector	OUTPUT	x1	3 Position Wire to Board Terminal Block	Through Hole 2.54mm	C49291873	C254128V-3P0G36	C49291873	C49291873	\$ 0.3968 x 1
Connector	DC power	x1	2 Position Wire to Board Terminal Block	Through Hole 2.54mm	C49291872	C254128V-2P0G36	C49291872	C49291872	\$ 0.2661 x 1
Switch	EN, BOOT	x2	Black Round Button SMD	4mmx3mmx2mm	C720477	TS-1088-AR02016	C720477	C720477	\$0.0555 x 1
IC	USB-C PD	x1	3V/5.5V 160uA -40°C~+105°C	2.6mmx2.6mmx0.5mm	C2678061	STUSB4500QTR	C2678061	C2678061	\$1.3770 x 1
IC	Encoder	x1	3V~3.6V 7mA -40°C~+150°C	5mmx6mmx1.75mm	C2655191	TLE5009E1000FUMA1	C2655191	C2655191	\$3.7725 x 1
IC	Microcontroller	x1	2.4GHz 3V~3.6V 97mA -40°C~+85°C	20.5mmx15.4mmx2.4mm	C2913206	ESP32-S3-MINI-1-N8	C2913206	C2913206	\$4.5968 x 1
IC	Motor driver	x1	4.5-35V 8A peak 200-kHz PWM -40°C~+125°C	5mmx7mmx1mm	C20345369	DRV8316CQRGRGFRQ1	C20345369	C20345369	\$6.6228 x 1
IC	Voltage regulator	x1	500kHz Buck Adjustable 2A 4.5V~28V	1.6mmx2.9mmx1mm	C191884	TPS54202DDDCR	C191884	C191884	\$0.2030 x 1
Mosfet	Power mosfet for power switching	x1	30V 40A	3.15mmx3.3mmx0.75mm	C49452333	STL9P3LLH6-TP	C49452333	C49452333	\$0.1990 x 1
Diode	VCC buss diode	x3	30V 5A	5mmx2.7mmx1mm	C50603	PMEG3050EP.115	C50603	C50603	\$0.4075 x 1
Diode	ESD Diode	x1	25VC Clamp ESD DIODE	1.6mmx3mmx1.4mm	C95343	ESDA25L	C95343	C95343	\$0.0826 x 1
Diode	ESD Diode	x3	38VC Clamp ESD DIODE	1.6mmx1mmx0.5mm	C19829649	ESDA25P35-1U1M	C19829649	C19829649	\$0.1098 x 1